

SHARAD Subsurface Interface Report

Product: r_0353601_001_ss19_700_a

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Detection Summary

Confidence: HIGH
Detection rate: 95.9% of traces
Coherent segments: 21
Mean interface depth: 100 range bins (3.74 us two-way)

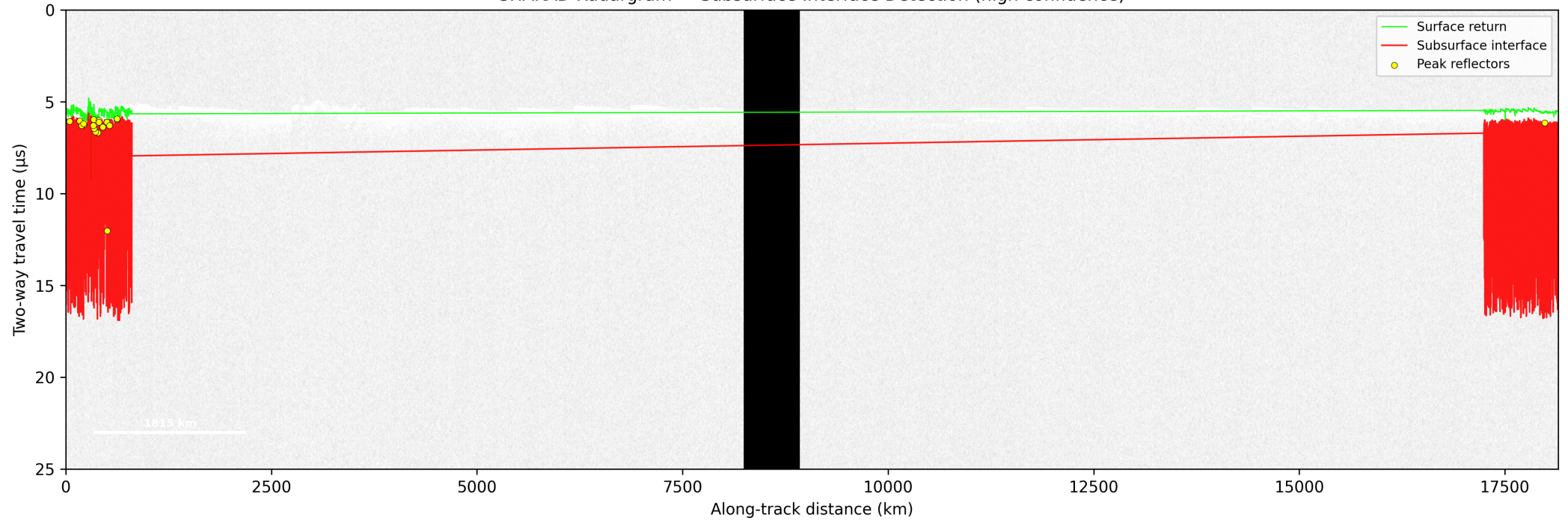
Depth Estimates

Material	er	Mean (m)	Min (m)	Max (m)
Vacuum	1.0	560.9	84.3	1680.7
Porous Ice	2.8	335.2	50.4	1004.4
Pure Water Ice	3.1	318.6	47.9	954.6
Basalt (low)	4.0	280.4	42.2	840.4
Basalt (mid)	5.0	250.8	37.7	751.6
Basalt (high)	6.0	229.0	34.4	686.1

HiRISE DTM used: DTEEC_038143_2205_055536_2205_A01

Annotated Radargram with Subsurface Interface

SHARAD Radargram — Subsurface Interface Detection (high confidence)



Interface Detection Mask



Scientific Interpretation

A laterally coherent subsurface reflector was detected with HIGH confidence.

Assuming pure water ice ($\epsilon_r = 3.1$), the mean interface depth is 318.6 m (range: 47.9-954.6 m).

Assuming dry basaltic regolith ($\epsilon_r = 4.0$), the mean depth is 280.4 m.

Dielectric back-calculation was not possible (insufficient DTM coverage or relief).